WHEN OBJECTIVE AMBIVALENCE LEADS TO SUBJECTIVE AMBIVALENCE FOR HEALTH BEHAVIORS: AN AFFECT-COGNITION MATCHING PERSPECTIVE

REINER NG WEI JIE

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Declaration

I hereby declare that this thesis is my original work and it has been written by me in its entirety. I have duly acknowledged all the sources of information which have been used in the thesis.

This thesis has also not been submitted for any degree in any university previously.

Reiner Ng Wei Jie

Date: 11/09/2020

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Summary

The extent to which people actually simultaneously hold opposite reactions to an attitude object (i.e., objective ambivalence) does not always lead people to psychologically experience conflicts in their attitudes (i.e., subjective ambivalence). Yet, subjective ambivalence has been established to impact important outcomes such as information processing and attitude-behavior consistency. As such, much research has examined factors that attenuate or enhance the relationship between objective and subjective ambivalence. However, this relationship has yet to be examined in the context of matching people's information focus to a topic's affect-cognition orientation. Because a matched focus should mean increased ability to process internal reactions, it should also increase the awareness of the extent to which opposite reactions are held. Thus, the current research hypothesizes that the relationship between objective and subjective ambivalence should be stronger under a matched focus to the topic compared to a mismatched focus. Furthermore, these effects should be stronger when people are more motivated to defend their attitudes. In Study 1, participants' objective ambivalence was measured before they were presented with information to induce a matched or a mismatched focus for a cognitive topic. They were then asked to report their subjective ambivalence. The results showed that objective ambivalence positively predicted subjective ambivalence in the matched condition, but not in the mismatched condition. Studies 2-3 extended these results by examining affective and cognitive topics, respectively, and by measuring participants' initial attitudes as a proxy for which the information's position would trigger the defense motivation. Results showed that as hypothesized, the effects of a

matched focus on the relationship between objective and subjective ambivalence were greater among those for whom the information triggered a defense motivation. Implications for our understanding of subjective ambivalence and affect-cognition matching will be discussed.

Keywords: ambivalence, affect, cognition, attitudes, matching

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Introduction

Attitudes are not simply about having *positive or negative* evaluations, with much research showing that attitudes can vary on different attitudinal strength and structural dimensions (Fabrigar et al., 2005; Krosnick & Petty, 1995). One well-researched attitudinal strength dimension is that of objective ambivalence. Specifically, objective ambivalence happens when a person simultaneously has both *positive and negative* evaluations toward the same target, resulting in different degrees of potential evaluative conflicts (Thompson et al., 1995). For example, receiving flu vaccination can confer protection but can be expensive; drinking sweetened drinks can feel delightful but it can harm your health. As people can face evaluative conflicts for a wide range of topics (e.g., abortion rights, political candidates, racial prejudices) and behaviors (e.g., junk food and meat consumption, pro-environmental behaviors), a better understanding of ambivalence can provide utility across many different settings (Armitage & Conner, 2000; Castro et al., 2009; Craig et al., 2002; Hass et al., 1992; McGraw et al., 2003; Sparks et al., 2001). For instance, objective ambivalence can result in an uncomfortable experience known as subjective ambivalence, consequently affecting many important outcomes such as information processing, attitude-behavior consistency, and attitude stability (Armitage, 2003; Clark et al., 2008b; Luttrell et al., 2016a; Nordgren et al., 2006). As such, much of the existing ambivalence research has focused on explicating the objective-subjective ambivalence relationship.

Although a growing body of research has examined factors that strengthen or attenuate the objective-subjective ambivalence relationship (e.g., Haddock et al., 2017; Newby-Clarke et al., 2002; Weng & DeMarree, 2019),

as far as it is known, none has examined the role of affective-cognitive information focus as a factor that influences this relationship. Since affective and cognitive information are important building blocks for attitudes (Eagly & Chaiken, 2007; Haddock & Zanna, 1999), knowing when an affective or a cognitive information focus can affect the correspondence between objective and subjective ambivalence can help manage important downstream outcomes. As such, the current research seeks to elucidate how affective-cognitive matching between the individual's information focus induced by the given information and the topic's orientation can influence the amount of subjective ambivalence experienced. Specifically, it is proposed that for an affective (cognitive) topic, having a matched focus on affect (cognition) would result in stronger objective-subjective ambivalence correspondence than when having a mismatched focus on cognition (affect). That is, having high (low) objective ambivalence should result in more (less) subjective ambivalence (see Figure 1).

Defining and Measuring Objective and Subjective Ambivalence

In the following sections, the different conceptualizations of ambivalence will be provided and distinguished from other seemingly similar constructs. Objective ambivalence refers to the simultaneous presence of strong positive and negative evaluations toward the same attitude object that people actually have (Jonas et al., 2000). Being an operative measure of attitude strength, objective ambivalence examines the direct mental processes (e.g., intensity) for the actual evaluative conflicts that individuals may be unaware of (Bassili, 1996; see Bassili, 2008). Two aspects of objective ambivalence are 1) similarity in the amount of positive and negative

evaluations and 2) moderately intense positive and negative evaluations (Hass et al., 1991). In the ambivalence literature, objective ambivalence has also been termed as “potential” ambivalence and “structural” ambivalence (e.g., Conner & Armitage, 2008; Refling et al., 2013).

Since ambivalence is theorized to be bi-dimensional instead of unidimensional (Cacioppo et al., 1997; Conner & Sparks, 2002), ambivalent individuals would simultaneously hold positive and negative evaluations instead of only holding either evaluation. As such, bipolar semantic differential scales cannot effectively measure objective ambivalence. For example, items like, “I am _____ about [attitude object]” with options ranging from “positive” to “negative” poorly distinguish ambivalent attitudes from indifferent or neutral attitudes (Klopfer & Madden, 1980). Critically, this is because they do not inform the degree of similarity and intensity of the positive and negative reactions respectively. As such, objective ambivalence is measured with unipolar Likert scales to capture the positive and negative evaluations separately (Thompson et al., 1995). For example, participants can be presented with a question that focuses on their favorability “Considering only the positive things about eating a low-fat diet in the future, how positive are those things?” as well as a separate question that focuses on their opposition “Considering only the negative things about eating a low-fat diet in the future, how negative are those things?” (e.g., Armitage & Conner, 2000). These positive and negative scores are then used to compute an index to inform the individual’s degree of objective ambivalence.

In comparison, subjective ambivalence is the negative unpleasant experience of conflict a person has when considering an attitude object

(Priester & Petty, 1996). Subjective ambivalence is considered as a meta-attitudinal measure of attitude strength that examines the meta-judgment of the conflict individuals perceive they experience (Bassili, 1996; see Bassili, 2008). That is, subjective ambivalence focuses on the negative conflicting experiences instead of the actual evaluative conflicts (van Harreveld et al., 2009). Subjective ambivalence is also referred to as "felt" ambivalence (e.g., Armitage & Arden, 2007). Typically, subjective experience of conflicts is measured with Likert scale items such as, "I feel _____ conflicted toward [attitudinal target]", with the choice of "not at all" to "completely" at the ends of the scale (Priester & Petty, 1996). Hence, to summarize, objective ambivalence refers to the structural conflict and measures the direct mental processes for the actual evaluative conflict, whereas subjective ambivalence refers to the experienced conflict and measures for the perceived amount of conflict experienced.

Distinguishing Objective Ambivalence from Other Inconsistency Constructs

At first glance, objective ambivalence might appear similar to other inconsistency-related constructs such as indifference and uncertainty despite being conceptually different. Specifically, while both indifferent (i.e., neither positive nor negative) and ambivalent (i.e., equally positive and negative) participants are equally likely to choose the midpoints of a bipolar scale (Kaplan, 1972), they have different attitudinal properties. That is, indifferent attitudes mean having weak to no positive and negative evaluations while ambivalent attitudes have at least some degrees of positive and negative evaluations (Cacioppo et al., 1997). Thus, holding ambivalent attitudes can result in different outcomes from indifferent attitudes. For example, despite

similar bipolar attitude scores, ambivalent individuals were more likely to smoke than indifferent individuals (Larsen & Cohen, 2009). More importantly, indifferent individuals are unlikely to experience subjective ambivalence given the lack of actual conflicting evaluations (Priester & Petty, 1996). Therefore, ambivalence should not be confused with indifference as they have different attitudinal properties and thus, can result in different outcomes.

Uncertainty, the subjective lack of confidence or conviction in one's attitudes, can also be distinguished from objective ambivalence (Gross et al., 1995, Tormala & Rucker, 2007). That is, having conflicting evaluations is not synonymous with low conviction in one's attitudes. For example, Clarkson et al. (2008) found that attitude uncertainty was unaffected by whether a person had learned consistent or conflicting information. Instead, it was influenced by the credibility of the information source. Thus, this showed that the two constructs are not interchangeable as the amount of uncertainty did not correspond to the amount of conflicting information learned. Moreover, considering both sides of an issue, which creates evaluative conflicts, was found to reduce attitude uncertainty instead of increasing it (Rucker et al., 2008). Hence, this further supports the distinction of the two constructs as attitude uncertainty did not increase with more evaluative conflicts, and instead showed a reduction. Thus, attitude uncertainty is distinct from objective ambivalence as it does not parallel the changes observed in objective ambivalence and can even change in the opposite direction.

Moderators of the Objective-Subjective Ambivalence Relationship

Given that subjective ambivalence can affect many important outcomes such as information seeking and processing, attitude-behavior

consistency, and attitude stability (Armitage, 2003; Clark et al., 2008b; Luttrell et al., 2016a; Nordgren et al., 2006), many studies have focused on examining its antecedents which includes objective ambivalence. In early ambivalence research, a high degree of objective ambivalence was assumed to result in more subjective ambivalence (e.g., Jonas et al., 1997; Maio et al., 1996). However, subsequent research that used different mathematical formulae to compute for objective ambivalence only found moderate correlations (*rs* = .16 to .52) between objective and subjective ambivalence (Lipkus et al., 2001; Priester & Petty, 1996). Put differently, this meant that objective ambivalence alone can only account for at most 27% of the variance in subjective ambivalence. Importantly, this also meant that despite being correlated, objective ambivalence and subjective ambivalence are independent constructs (e.g., Lipkus et al., 2001; Newby-Clarke et al., 2002; Priester & Petty, 1996). That is, a person need not always experience a sense of conflict even when they actually hold a high degree of conflicting reactions toward an attitude object. Thus, although a moderate degree of correspondence between objective and subjective ambivalence exists, the two constructs are not interchangeable and need not always lead to each other.

Indeed, according to past research, several situational factors can influence the objective-subjective ambivalence relationship. That is, these factors influence whether objective ambivalence remains dormant or becomes salient to result in subjective ambivalence. For example, Newby-Clarke et al. (2002) found that simultaneous awareness of both sides of an issue increases the degree of correspondence between objective and subjective ambivalence. Other situations, such as a forced decision between binary options, can also

increase the objective-subjective ambivalence correspondence as it increases a person's awareness of the benefits and shortcomings of either option (Itzchakov & van Harreveld, 2018; van Harreveld et al., 2009). At the cultural level, because East Asians tend to interpret their evaluative conflicts as being informative instead of being inconsistent, they are less likely to experience subjective ambivalence (Ng et al., 2012; Wang et al., 2016). That is, East Asians are less aware of their conflicting evaluations because they tend to view the inconsistencies to be compatible rather than conflicting. Thus, this results in weaker objective-subjective ambivalence relationship as they experience lesser discomfort from the evaluative conflicts.

Dispositional factors can also influence the objective-subjective ambivalence relationship. Individuals who are prepared for change showed a stronger objective-subjective ambivalence relationship (r = .63 vs .12) compared to those unprepared for change (Armitage & Arden, 2007). This could be due to prepared individuals having more active considerations of their evaluative conflicts so that they do not risk increasing their experienced discomfort (van Harreveld et al., 2009). Preference for consistency also matters as those with a high preference for consistency are more likely to experience conflict when evaluative conflicts are salient compared to those with a low preference (Newby-Clarke et al., 2002).

Considering these past findings together, a common theme among them is that the awareness of the evaluative conflicts is a factor that influences the objective-subjective ambivalence relationship. Extending this line of research, this research submits that reading topic-related information that is matched to the topic's affective-cognitive orientation should also increase the

awareness of the conflicting evaluations about the topic, thereby influencing the correspondence between objective and subjective ambivalence.

Affective-Cognitive Matching

Past research has highlighted the usefulness of affective-cognitive matching where a matched focus (e.g., cognitive focus to cognitively oriented topic) usually produce better outcomes compared to a mismatched focus (e.g., affective focus to cognitively oriented topic) (Giner-Sorolla, 2004; van den Berg et al., 2006). For example, some outcomes include greater reading interests, faster reading speed, and more positive judgments of the information under a matched condition compared to a mismatched condition (e.g., Keer et al., 2013; See et al., 2013a). One explanation for this is that under a matched focus, people have greater processing fluency for the information that matches than mismatches the activated focus (Mayer & Tormala, 2010). Specifically, this greater processing fluency has been attributed to the increased accessibility and recall of related information that matches the induced focus (Giner-Sorolla, 2004; Lee & Aaker, 2004). Put differently, the greater processing fluency from a matched focus reflects enhanced ability to process internal reactions to the information being read and recalled.

Affective-Cognitive Focus

Following prior research, the affective-cognitive focus here refers to the situational inductions that influence whether a person focuses more on their feelings and emotions or their thoughts and beliefs when forming their attitudes (van den Berg et al., 2006). The goal of such situational inductions is to influence only the affective-cognitive focus without impacting the overall attitudes. That is, the assumption is that two individuals can differ in how

affective or cognitive their focus is, while arriving at the same attitude. One such situational induction uses instructions and statements. For example, Farley and Stasson (2003) induced a cognitive focus by instructing participants to think about blood donation ("indicate your beliefs about blood donation"), and induced an affective focus by asking them their feelings about blood donation ("indicate how blood donation makes you feel"). Other studies that have used similar approaches include Millar and Tesser (1986) which instructed participants to focus on affective versus cognitive reactions respectively. Whereas Mayer and Tormala (2010) did so by giving "I feel" versus "I think" statements to participants. Similarly, an affective-cognitive focus can also be situationally induced by providing participants with affective or cognitive information. For example, van den Berg et al. (2006) and Giner-Sorolla (2004) provided participants with affectively oriented words such as feelings, emotions, exciting, and frightening to induce an affective focus. In contrast, a cognitive focus was induced using cognitively oriented words such as thinking, rational, efficient, and advanced. Hence, these studies support that an affective or a cognitive focus can be induced situationally.

Topic's Affective-Cognitive Orientation

A topic's affective-cognitive orientation refers to the affective or cognitive attitudes and reactions that are primarily associated with the topic across individuals (Crites et al., 1994; Eagly et al., 1994; Lawton et al., 2009). Various origins for a topic's affective-cognitive orientation have been proposed. For instance, objects and topics that serve to fulfill self-expression (e.g., flags) are affectively oriented whereas those that serve utilitarian needs (e.g., coffee) are cognitively oriented (Maio & Olson, 2000; Shavitt, 1990). As

another example, topics that normatively raise collectivistic concerns (e.g., clothes iron, laundry detergent) are more cognitively oriented whereas those that are personal or private (e.g., toothbrush, running shoes) are more affectively oriented (Hong & Chang, 2015; Shavitt et al., 2009). Of relevance to the current research, it is suggested that health behaviors can also be identified and categorized according to their affective-cognitive orientation. For example, harmful behaviors such as smoking tend to be more affectively oriented while healthy behaviors such as consuming fruits and vegetables are more cognitively oriented (Janssen et al., 2012; Lawton et al., 2009). One possible explanation for this is that harmful health behaviors tend to create stronger emotional reactions from the detrimental impact it has on a person's well-being (Leventhal & Cameron, 1987; Loewenstein et al., 2001). In contrast, healthy behaviors allow for the consideration of the instrumental benefits it provides as it does not threaten and instead promote a person's well-being (Millar & Millar, 1993). Nonetheless, regardless of the origins of the pre-existing affective-cognitive orientations of topics, prior affective-cognitive matching research (e.g., Crites et al., 1994; Eagly et al., 1994) has shown that the affective-cognitive orientation distinction is a meaningful way to categorize and identify topics.

Greater Awareness of Ambivalence from Matched Focus

Applying affective-cognitive matching to the ambivalence context, a matched focus should result in a stronger objective-subjective ambivalence relationship compared to a mismatched focus. That is, when reading information about a topic, a person evaluative conflicts about the topic should be more salient when the affective-cognitive nature of the information matches

the topic's orientation compared to when it mismatches. This is because matching would increase the perceived ability to process their conflicting reactions or to access or recall both supporting and opposing information and reactions in which the attitudes toward the topic are based on (Giner-Sorolla, 2004; See et al., 2013b). As such, if a person does hold conflicting information about the topic, then the greater accessibility, recall, or processing of the different pieces of conflicting information should translate into more subjective ambivalence as seen in earlier research (e.g., Newby-Clarke et al., 2002). Conversely, if a person holds mainly one-sided information about the topic, then the greater accessibility and recall would only bring univalent information to awareness, translating to even lesser subjective ambivalence compared to those under a mismatched focus.

Influence of Initial Attitudes

Furthermore, this affective-cognitive matching effect for the objective-subjective ambivalence relationship should also depend on the individual's initial attitudes. Initial attitudes are important as it informs the person's position about the attitude object and guides their interactions with it (Eagly & Chaiken, 2007; Neumann et al., 2004; Trafimow, 1996). Hence, when induced with a matched focus to the topic's orientation, the increased perceived ability to attend to their evaluations about the topic should motivate more deliberations. However, people may not always want to deliberate, and may only do so when their held position is challenged (Chaiken et al., 1989).

Heuristic-Systematic Model

A possible outcome when initial attitudes are considered would be a stronger objective ambivalence × information focus interaction pattern among

individuals with attitudes that are relatively inconsistent to the information, compared to those with consistent attitudes. That is, a stronger interaction effect is expected for people who view the focus induction information to be discrepant with their attitudes compared to those who view it to support their attitudes. Based on the Heuristic-Systematic Model (HSM), being presented with information that is discrepant with one's attitudes can elicit defense-motivated processing which promotes information scrutiny (Chen & Chaiken, 1999). This is based on the sufficiency principle in the HSM where people generally want to hold their attitudes with certainty, and when their original attitudes are threatened by discrepant information, they would process the information carefully to attempt to regain confidence (Chaiken et al., 1989). Past studies support this as systematic processing usually occurs under perceived hostile cues, while heuristic processing usually happens under supportive cues (Giner-Sorolla & Chaiken, 1997; Liberman & Chaiken, 1992). Hence, individuals' initial attitudes would act as a proxy for whether the information given to induce the affective-cognitive focus would trigger the defense motivation for careful processing. Taken together, individuals with inconsistent attitudes to the information are expected to show a stronger affective-cognitive matching effect on their objective-subjective ambivalence relationship. That is, a matched focus should give a stronger objective-subjective ambivalence relationship compared to a mismatched focus.

On the other hand, having consistent attitudes to the information should not trigger the defense motivation (Giner-Sorolla & Chaiken, 1997). The sufficiency principle suggests that individuals prefer to expend minimal cognitive efforts and adopt heuristic processing when there is no need to

defend their attitudes. As such, given the lack of defense motivation, individuals are unlikely to process the information deeply which would not enhance the saliency of their objective ambivalence. Hence, those with consistent attitudes to the information should not show the objective ambivalence × information focus interaction for their subjective ambivalence. This is because even though the matched focus should lead to greater ability to retrieve both attitude-consistent and attitude-inconsistent information than the mismatched focus, the lower motivation to consider their conflicting evaluations means that the impact of matched focus on magnifying the effects of objective ambivalence on subjective ambivalence is attenuated.

Underlying mechanisms aside, before the influence of initial attitudes can be examined, it will be important to first demonstrate the postulated affective-cognitive matching pattern. That is, under a matched focus to the topic's affective-cognitive orientation, individuals should experience more (less) subjective ambivalence compared to those with a mismatched focus, if they have a high (low) degree of objective ambivalence.

The Present Research

In this present research, the primary goal is to demonstrate that the interaction between objective ambivalence and the matched information focus to the topic's orientation can influence the objective-subjective ambivalence relationship. By showing the above, it would support the postulation that having a matched focus to the topic's affective-cognitive orientation would result in a stronger objective-subjective ambivalence relationship. Hence, as explained above, it is hypothesized that for those with a low degree of objective ambivalence, having a matched focus will result in less subjective

ambivalence than a mismatched focus. On the other hand, for those with a high degree of objective ambivalence, having a matched focus will result in a greater amount of subjective ambivalence than a mismatched focus. In other words, under a matched focus, objective ambivalence should better predict subjective ambivalence compared to a mismatched focus.

Since no studies have examined the interaction effects from matching affective-cognitive information focus and topic's orientation with objective ambivalence on the subjective ambivalence experience, Study 1 was a preliminary investigation to verify the hypotheses. Participants reported their objective ambivalence and were induced with either a matched or a mismatched information focus for a cognitively oriented topic (receiving flu vaccination) before reporting their subjective ambivalence. Study 2 then examined these effects for an affectively oriented topic (drinking sweetened drinks). Also, Studies 2 and 3 examined how initial attitudes could further influence these effects. Study 3 was performed using the same cognitively oriented topic from Study 1 to ensure that the results which considered the influence of initial attitudes were generalizable across affectively and cognitively oriented topics.

Study 1: Receiving Flu Vaccination

Flu vaccination is the examined cognitive topic as it is a health-promoting behavior with many benefits including improved immunity, reduced influenza risks, and reduced illness severity (Centers for Disease Control and Prevention, 2019). Moreover, as a health-promoting behavior, flu vaccination rarely threatens individuals' wellbeing which allows them to process the utility of the proclaimed benefits (Millar & Millar, 1993).

Supporting this, Prislin et al. (1998) found that vaccination status was mediated by cognitive factors such as beliefs for diseases protection and immunity. Similarly, other beneficial behaviors, such as condom use for sexual diseases prevention, taking vitamins for immunity, and teeth flossing are also better predicted by cognitive attitudes than affective attitudes (Glasman & Albarracín, 2006; Lawton et al. 2009). Indirect evidence also supports flu vaccination to be a cognitive topic since information that focused on the instrumental benefits of vaccinations did improve vaccination intentions (e.g., Abhyankar et al., 2008; Gerend & Shepherd, 2007).

However, despite being safe and relatively accessible in developed countries, many countries such as Singapore and United States still lack herd immunity due to poor vaccination rate (e.g., 20% to 40%) in their populations (Ang et al., 2017; Centers for Disease Control and Prevention, 2019). Since the upside of having a well-vaccinated society can save $USD16.3 billion annually in the United States alone (Molinari et al., 2007), there is much utility to examine how subjective ambivalence for flu vaccination behavior can be managed using an affective-cognitive matching approach (e.g., Kim et al., 2019). Taken together, since flu vaccination is a cognitive topic, it is proposed that the matched cognitive focus should give a stronger objective-subjective ambivalence correspondence compared to the mismatched affective focus.

Method

Participants and Design

Participants were 136 (M_{age} = 20.18, *SD* = 1.55, 75.70% female) university students at the National University of Singapore who participated either for partial course credit or SGD5. Power analyses suggested that *N* =

133 was required for 80% power to detect the two-way interaction of objective ambivalence × affective-cognitive information focus with an effect size as small as f^2 = .06 (Faul et al., 2009). Participants were randomly assigned to either an affective or cognitive information focus. The outcome variable, subjective ambivalence was measured as a continuous variable.

Procedures

Each session had up to six participants in individual cubicles completing the computerized study presented using the Medialab software (Jarvis, 2014). Before the study, participants were briefed that there were two unrelated studies. This was done to manage their expectations and the possible response biases (Dunn, 2013). For example, participants may notice the affective-cognitive information focus that is being induced by the given information and change their responses intentionally based on what they perceive the hypotheses to be. Different introduction screens also appeared before the start of the ostensibly different studies to further reduce participants' suspicion that the two studies were related.

On the computer screen, the first study appeared as an opinion survey where participants were briefed to provide their views on receiving yearly flu vaccinations. Negative and positive evaluation items were presented in a randomized manner (Refling et al., 2013). These responses toward flu vaccination were then used to compute the participants' degree of objective ambivalence towards receiving yearly flu vaccination.

Then, a different introduction screen indicated the start of the supposedly different study. Participants were told that they were randomly

assigned to read health campaign information that appeared recently in the media. However, all participants would receive information that either induced an affective or a cognitive focus that outlined the benefits of receiving yearly flu vaccinations. After reading, participants reported their subjective ambivalence towards receiving yearly flu vaccination and completed the demographic items before being debriefed.

Predictor Variables

Degree of Objective Ambivalence. The degree of objective ambivalence was assessed before the participants were induced with an affective or a cognitive focus. Responses were provided on a scale ranging from 1(*strongly disagree)* to 7(*strongly agree).* Specifically, three positive (M = 3.645, SD = 1.200, α = .665) and three negative (M = 3.885, SD = 1.524, α = .761) unipolar items adapted from past research were used to measure the participants' objective ambivalence (Carfora et al., 2016; Crites et al., 1994; Smith & Nosek, 2011). These positive/negative items include, "I feel *happy/bored* about receiving yearly flu vaccination, it can feel *delightful/sad* to receive yearly flu vaccination, and it feels *exciting/disgusting* to receive yearly flu vaccination". The responses for the positive and negative items were averaged separately. Then, the two averaged values were used to compute for an objective ambivalence index towards receiving yearly flu vaccination (see Appendix A for objective ambivalence items for all studies).

The index (M = 2.015, SD = 1.662) was computed using the formula referenced from Thompson et al. (1995): $[(P + N) / 2] – |P – N|$. "P" and "N" are the average of positive and negative responses. Scores could range between -2 to +7, with a larger positive score suggesting greater similarity and

intensity among the positive and negative responses toward receiving yearly flu vaccination (i.e., more objective ambivalence).

Affective-Cognitive Information Focus. Participants were then induced with either an affective or a cognitive information focus regarding the benefits of receiving yearly flu vaccinations. In the *affective focus condition*, the information used emotional adjectives to describe the positive outcomes resulting from receiving flu vaccinations. For example, "most people are happy and excited to learn that receiving yearly flu vaccination greatly reduces their risk of getting sick with flu and flu-associated hospitalisation." In the *cognitive focus condition*, the information used cognitively oriented adjectives to describe the safety and the benefits of flu vaccinations. For example, "most people know that a yearly flu vaccination is safe, and greatly reduces their risks of getting sick with flu and flu-associated hospitalisation". An equal number of advantages were described, and the passages were similar in length (see Appendix B for information used).

Outcome Variable

Subjective Ambivalence. Following the affective-cognitive focus induction, participants reported their subjective ambivalence towards receiving yearly flu vaccination on the scale adapted from Priester & Petty (1996) ranging from 1(*not at all*) to 9(*totally*). The three Likert items used were, "to what extent do you feel conflicted/mixed/undecided about receiving yearly flu vaccinations?" The responses were averaged, with higher scores suggesting greater subjective ambivalence (M = 3.583, SD = 1.975, α = .945).

Results and Discussion

Table 1 shows the zero-order correlations among the predictors and the outcome variable. Consistent with the assumption that objective and subjective ambivalence are different constructs and need not always correlate, objective and subjective ambivalence did not correlate (r = .092, p > .05).

Outcome Variable: Subjective Ambivalence

A hierarchical multiple regression was conducted with subjective ambivalence as the outcome variable (see Table 2). Participants' objective ambivalence towards receiving yearly flu vaccination and affective-cognitive information focus (dummy coded as: 0 = cognitive, 1 = affective) were entered first, before entering the objective ambivalence × affective-cognitive information focus interaction. In this and all subsequent regression analyses, effects were interpreted in the first model that they appear (Cohen et al., 2003). In addition, all continuous predictor variables were centered. Participants with a lower or higher degree of objective ambivalence were computed at one standard deviation below and above their means (Aiken & West, 1991).

The analyses revealed no main effect of objective ambivalence, B = 0.119, t(133) = 1.140, p = .256, 95% CI: [-0.087, 0.326], r_{partial} = .098. There was also no effect of the affective-cognitive information focus, B = -0.173, t(133) = -0.500, p = .618, 95% CI: [-0.857, 0.511], r_{partial} = -.043. Of importance, a significant objective ambivalence × affective-cognitive information focus interaction was observed, B = -0.690, t(132) = -3.410, p = .001, 95% CI: [-1.090, -0.290], r_{partial} = -.285. In order to test the hypothesis that a matched information focus to the topic's orientation can result in

stronger objective-subjective ambivalence relationship, the two-way interaction was decomposed by the degree of objective ambivalence. The results revealed that as expected, the tendencies of objective ambivalence to predict subjective ambivalence were stronger under a matched focus than a mismatched focus.

Specifically, low objective ambivalence participants with matched topic orientation and information focus reported even lower subjective ambivalence compared to those with mismatched topic orientation and information focus, B = 1.002, $t(132)$ = 2.092, p = .038, 95% CI: [0.055, 1.950], $r_{partial}$ = .179. At the same time, as expected, for high objective ambivalence participants, a matched topic orientation and information focus resulted in more subjective ambivalence than a mismatched topic orientation and information focus, B = -1.297, $t(132)$ = -2.769, p = .006, 95% CI: [-2.224, -0.370], $r_{partial}$ = -.234 (see figure 2). Hence, the hypothesis where having a matched information focus to the topic's affective-cognitive orientation can result in stronger objective-subjective ambivalence relationship compared to a mismatched focus is supported.

The primary purpose of Study 1 was to examine if the matching between individuals' affective-cognitive information focus (as induced by the exposure to affective or cognitive information) and a topic's pre-existing orientation would interact with objective ambivalence to influence the amount of subjective ambivalence experienced. Supporting the hypothesis, participants focusing on cognition that have low objective ambivalence reported even lower subjective ambivalence than those focusing on affect. On the other hand, those with high objective ambivalence reported more

subjective ambivalence for a cognitively oriented topic. Based on past research (e.g., Giner-Sorolla, 2004; van den Berg et al., 2006), participants induced with a mismatched affective focus could have lower accessibility for relevant information to a cognitive topic, resulting in lower awareness of their conflicting evaluations. As such, high objective ambivalence individuals who received a mismatched focus did not experience as much subjective ambivalence as those who received a matched focus.

Additionally, the findings also hints that just as a matched focus can result in a stronger objective-subjective ambivalence relationship to promote or reduce attitude-behavior consistency (Armitage & Conner, 2000), a mismatched focus can also serve similar functions if used in the reversed manner. That is, to promote a behavior, a mismatched focus can be induced for those with high objective ambivalence to reduce their experienced conflicts for increased likelihood of behavior. Conversely, to reduce a behavior, a mismatched focus can be induced for low ambivalent individuals to maintain or to increase their subjective ambivalence, thereby reducing the likelihood of behavior (van Harreveld et al., 2015).

Study 2: Drinking Sweetened Drinks

Study 2 extended the findings from the previous study in a few ways. First, it was of interest to show that the interaction pattern observed in Study 1 was not unique to cognitively oriented topics. That is, Study 2 wanted to rule out the alternative explanations where the interaction effects observed in Study 1 could be attributed to other potential influences arising from a cognitive topic. For example, a cognitive topic might naturally elicit more deliberate judgment on the probable utility of the information before it is integrated into

a person's evaluations (Fishbein & Ajzen, 1975; Olson & Kendrick, 2008). As such, cognitive topics might naturally lead to more deliberations that can result in similar interaction effects from Study 1. Hence, Study 2 aimed to demonstrate that the affective-cognitive matching effects would also hold for an affectively oriented topic. Thus, if the findings from Study 1 are accounted for by other extraneous influences arising from cognitive topics (e.g., greater deliberations) instead of the affective-cognitive matching effects, then the interaction pattern in Study 2 should again occur among those focusing on cognition. However, if the findings are accounted for by the hypothesized affective-cognitive matching effects, then Study 2 should show a different pattern from that of Study 1. That is, the correspondence between objective and subjective ambivalence would be stronger among those randomly assigned to focus on affect than others that are assigned to focus on cognition.

Drinking Sweetened Drink

Drinking sweetened drinks is the examined affective topic as it is a harmful health behavior which can result in many costly physiological and psychological repercussions including the increased risks for type II diabetes, cardiovascular issues, and mental distresses (Lien et al., 2006; Micha et al., 2017; Wilde et al., 2019). Specifically, harmful health behaviors are more affectively oriented as feelings and emotions are posited to play a more consequential role in making decisions for risky behaviors compared to beliefs and knowledge (Loewenstein et al., 2001). For example, in smoking behaviors, affective attitudes rendered cognitive attitudes non-significant after they were added to predict smoking intentions in adolescents (Lawton et al., 2007). In another study, Lawton et al. (2009) found that unlike healthy

behaviors (e.g., vitamin usage, fruits and vegetable consumption), harmful health behaviors (e.g., binge drinking, smoking) were better predicted by affective attitudes compared to cognitive attitudes. One explanation for this is that thinking or acting on harmful behaviors does not enhance one's health and could instead bring to awareness the dangers and risks in which their behaviors have on their health (Millar & Millar, 1993). As such, behaviors that threaten individuals' well-being are likely to generate prominent emotional concerns (e.g., repercussions of harmful behaviors). Given that several countries such as the United States and Spain have even resorted to sugar tax to mitigate the overconsumption of sweetened drinks (Lee et al., 2020; Royo-Bordonada et al., 2019), there is much urgency to manage this health behavior.

In addition, as Study 1 obtained support for the postulated affective-cognitive matching pattern, it allows for Study 2 to further examine the influence of initial attitudes on this interaction effect. Since the information seeks to reduce sweetened drink consumption, those with inconsistent attitudes (i.e., those who support drinking sweetened drinks) are expected to show the objective ambivalence × information focus interaction effect but not those with consistent attitudes (i.e., those who oppose drinking sweetened drinks). That is, only those with inconsistent attitudes to the information would have a stronger objective-subjective ambivalence relationship under a matched information focus to the topic orientation compared to a mismatched focus, but not those with consistent attitudes to the information.

Method

Participants and Design

After excluding eight participants who failed at least one of the three

attention checks (Niessen et al., 2016), the final sample consisted of 272 (M_{age} = 34.88, *SD* = 11.75, 45.60% female) individuals from the United States recruited from the Amazon M-Turk platform (Buhrmester et al., 2011). Each participant was compensated monetarily for their participation. Power analyses revealed that *N* = 247 were required to provide 80% power to detect the hypothesized three-way interaction of initial attitudes × objective ambivalence × affective-cognitive information focus with an effect size as small as f^2 = .06 (Faul et al., 2009). Hence, the current sample has enough statistical power to detect the hypothesized three-way interaction. Participants were randomly assigned to either an affective (*N* = 129) or cognitive information focus (*N* = 143). Subjective ambivalence is the outcome variable.

Procedures

The procedures were similar to Study 1 except that participants were recruited online on the Amazon M-Turk platform. Specifically, each participant completed two supposedly unrelated studies that were presented using the Qualtrics platform. In the first study, participants were informed that they had to give their views regarding drinking sweetened drinks. Positive and negative evaluation items were used to capture participants' responses to compute their objective ambivalence towards drinking sweetened drinks. Lastly, participants responded to items examining their initial attitudes toward drinking sweetened drinks. The positive and negative evaluation items and the initial attitudes items were presented together as a set.

After seeing a separate screen that signaled the start of the ostensibly separate study, participants were briefed that they were randomly assigned to read health campaign information taken from the media in the United States.

This time, the information sought to reduce than to promote the targeted behavior. That is, participants were induced with an affective or a cognitive information focus that described the risks and dangers of drinking sweetened drinks (see Appendix B for information used). Then, participants reported their subjective ambivalence toward drinking sweetened drinks. After the demographic items were completed, participants were debriefed.

Predictor Variables

Objective Ambivalence. Similar to Study 1, three positive (M = 4.074, SD = 1.746, α = .915) and three negative unipolar items (M = 4.966, SD = 1.106, α = .549) using different adjectives tapped for the participants' positive and negative responses on sweetened drinks. For example, "drinking sweetened drinks is *valuable for your health/creates a lot of challenges*, it is *wholesome/unnecessary* to drink sweetened drinks, drinking sweetened drinks is *beneficial/useless if you do not improve other areas of your health*". These responses were used to provide an objective ambivalence index using the same computation method from Study 1 (M = 2.995, SD = 2.316). A higher index score meant more objective ambivalence towards drinking sweetened drinks.

Initial Attitudes. Participants initial attitudes (M = 3.934, SD = 1.165, α = .782) toward drinking sweetened drinks were measured with six scale items adapted from past research (Crites et al., 1994; Smith & Nosek, 2011) that ranged from 1(*totally disagree*) to 7(*totally agree*). The items were presented in a randomized order. For example, "To what extent are your attitudes toward drinking sweetened drinks *positive/ negative?*" Their responses were recoded and then averaged, with higher score suggesting a more positive attitude.

Affective-Cognitive Focus. After reporting their attitudes, following prior research that induced affective-cognitive focus (e.g., van den Berg et al., 2006), participants were induced with either an affective or cognitive information focus that described the negative outcomes of drinking sweetened drinks. For example, the *affective focus* condition used affectively oriented adjectives to describe the negative emotional outcomes such as feeling more unhappy after the sugar high ends. On the other hand, the *cognitive focus* condition used cognitively oriented adjectives to describe the negative instrumental outcomes such as how drinking sweetened drinks can increase the risks for many health diseases. Thereafter, participants reported their subjective ambivalence.

Outcome Variable

Subjective Ambivalence. The same subjective ambivalence measure was adapted from Study 1. Participants indicated their subjective ambivalence towards drinking sweetened drinks on three items. Their responses were averaged, with higher score suggesting greater amount of ambivalence experienced (M = 5.300, SD = 2.345, α = .935).

Results and Discussion

Table 3 shows the zero-order correlations among the predictors and the outcome variable. In this study, objective and subjective ambivalence did correlate positively (r = .669, p < .01). There was a positive correlation between initial attitudes and objective ambivalence (r = .333, p < .01) and also subjective ambivalence (r = .309, p < .01).

Outcome Variable: Subjective Ambivalence

A hierarchical multiple regression was conducted with participants' subjective ambivalence as the outcome variable. Participants' initial attitudes, degree of objective ambivalence, and affective-cognitive information focus (dummy coded as: 0 = cognitive, 1 = affective) were entered first. Then, all possible two-way interactions were included before entering the three-way interaction term (i.e., initial attitudes × objective ambivalence × affective-cognitive information focus). The data analysis procedures remained identical to that of Study 1, with the addition of participants' initial attitudes. Similarly, having more negative or positive attitudes about drinking sweetened drinks were computed at one standard deviation below and above the initial attitudes mean (Aiken & West, 1991).

The analysis revealed that more objective ambivalence meant more subjective ambivalence, B = 0.645, $t(268)$ = 13.336, p < .001, 95% CI: [0.550, 0.741], $r_{partial}$ = .632. In addition, having more positive initial attitudes showed more subjective ambivalence, B = 0.197, $t(268)$ = 2.046, p = .042, 95% CI: [0.007, 0.386], $r_{partial}$ = .124. However, no effect of affective-cognitive information focus was observed, B = 0.147, $t(268)$ = 0.695, p = .488, 95% CI: [-0.269, 0.563], $r_{partial}$ = .042.

There was only one significant two way interaction of initial attitudes × objective ambivalence, B = -0.223, $t(265)$ = -4.858, p < .001, 95% CI: [-0.313, -0.133], $r_{partial}$ = -.286.[1] The remaining two-way interactions were not significant. That is, the initial attitudes × affective-cognitive information focus interaction was not significant, B = -0.320, $t(265)$ = -1.678, p = .095, 95% CI: [-0.696, 0.055], $r_{partial}$ = -.103. Likewise, the objective ambivalence

× affective-cognitive information focus interaction was also not significant, B = 0.077, t(265) = 0.821, p = .412, 95% CI: [-0.107, 0.260], $r_{partial}$ = .050.

More importantly, a significant three-way interaction was observed, B = 0.243, t(264) = 2.644, p = .009, 95% CI: [0.062, 0.423], $r_{partial}$ = .161. To examine if the initial attitudes mattered, this interaction was decomposed based on whether individuals have positive or negative attitudes toward drinking sweetened drinks. Among those with positive attitudes, which meant that they had inconsistent attitudes to the information, there was a marginal trend where more objective ambivalence meant more subjective ambivalence, B = 0.183, t(264) = 1.826, p = .069, 95% CI: [-0.014, 0.379], $r_{partial}$ = .112. However, there was no effect of affective-cognitive information focus, B = 0.169, t(264) = 0.499, p = .618, 95% CI: [-0.498, 0.836], $r_{partial}$ = .031. Of most importance, a significant two-way interaction was observed, B = 0.400, t(264) = 2.611, p = .010, 95% CI: [0.098, 0.701], $r_{partial}$ = .112. Similar to the interaction pattern from Study 1, among those with low objective ambivalence, a matched topic orientation and information focus resulted in marginally lesser subjective ambivalence than a mismatched topic orientation and information focus, B = -0.756, t(264) = -1.780, p = .076, 95% CI: [-1.593, 0.080], $r_{partial}$ = -.109. Conversely, for those with high objective ambivalence, a matched topic orientation and information focus resulted in more subjective ambivalence than a mismatched topic orientation and information focus, B = 1.094, t(264) = 1.997, p = .047, 95% CI: [0.015, 2.173], $r_{partial}$ = .122 (see figure 3a). Hence, this interaction pattern aligns with what one would expect from the interaction between the matching of affective-cognitive information focus and topic's orientation with the degree of objective ambivalence. That

is, a stronger objective-subjective ambivalence relationship occurred in those with a matched affective focus compared to a mismatched cognitive focus.

However, a different pattern was observed for those with negative attitudes, which meant that they had consistent attitudes to the information. For this group, more objective ambivalence meant more subjective ambivalence, B = 0.944, $t(264)$ = 10.447, p < .001, 95% CI: [0.766, 1.122], $r_{partial}$ = .541. But there was no effect of affective-cognitive information focus, B = -0.352, $t(264)$ = -0.809, p = .419, 95% CI: [-1.211, 0.506], $r_{partial}$ = -.050. More importantly, there was no significant interaction between objective ambivalence and information focus, B = -0.166, $t(264)$ = -1.275, p = .203, 95% CI: [-0.422, 0.090], $r_{partial}$ = -.078 (see figure 3b). In other words, when participants have attitudes consistent with the information, their subjective ambivalence mainly reflected their degree of objective ambivalence. Hence, the results showed that initial attitudes do matter as only individuals with attitudes inconsistent to the information showed the interaction pattern from Study 1.

The purpose of Study 2 was to ascertain if the findings from Study 1 would be supported for an affective topic and if individuals' initial attitudes would influence the interaction effects. Of relevance to the two objectives, the expected stronger objective-subjective ambivalence relationship in those with a matched focus did occur for an affective topic. That is, for those focusing on affect (versus focusing on cognition), having low objective ambivalence resulted in even lower subjective ambivalence. At the same time, having high objective ambivalence resulted in more subjective ambivalence. As such, the results do not support that the interaction pattern observed in Study 1 are due

to alternative explanations such as a greater inherent motivation to deliberate arising from a cognitive topic (Fishbein & Ajzen, 1975; Olson & Kendrick, 2008). More importantly, this would mean that the affective-cognitive matching effects are supported and could be generalized to both affective and cognitive topics. Moreover, individuals' initial attitudes can influence the interaction effects as the effects were stronger in those who were motivated to defend their attitudes (i.e., those with inconsistent attitudes to the information).

Study 3: Receiving Flu Vaccination

In Study 3, the same cognitive topic from Study 1 was examined to replicate the observed pattern from Study 2. This was done to give greater confidence that the findings considering initial attitudes can be generalized to both affective and cognitive topics.

Method

Participants and Design

Participants were 290 (M_{age} = 21.53, *SD* = 2.19, 69.30% female) university students at the National University of Singapore who participated in return for a SGD5 reimbursement. Hence, there was enough statistical power to detect the three-way interaction of initial attitudes × objective ambivalence × affective-cognitive information focus with an effect size as small as f^2 = .06. Similar to earlier studies, participants were randomly assigned to either an affective or cognitive information focus. Subjective ambivalence was measured as the outcome variable.

Procedures

The procedures were similar to that of Study 1 with some changes.

Each session had up to six participants that completed the computerized study presented using the Medialab software. On the computer screen of the first study, they were informed to give their opinions toward receiving yearly flu vaccinations. The same positive and negative evaluation items from Study 1 were used to capture participants' responses required for computing their objective ambivalence towards flu vaccination. Following the positive and negative items, participants also responded to items examining their initial attitudes toward flu vaccination.

Participants were then shown a separate study screen which briefed them that they were randomly assigned to read health campaign information taken recently from the media. Likewise, they receive information that either induced an affective or a cognitive information focus outlining the benefits of receiving yearly flu vaccination. Following this, participants reported their subjective ambivalence towards receiving yearly flu vaccination. After the demographic items were completed, participants were debriefed.

Predictor Variables

Objective Ambivalence. Following Studies 1 and 2, a total of three positive (M = 5.426, SD = 1.183, α = .898) and three negative unipolar items (M = 3.975, SD = 1.220, α = .540) tapped for the participants' positive and negative responses toward receiving yearly flu vaccination. The items include "I believe it is *wise/unnecessary* to get flu vaccinations, going for flu vaccination is *valuable* for your health/is *useless* if you do not improve other areas of your health, maintaining immunity against flu is *beneficial*/going for flu vaccination takes a long time and creates a lot of *hassle*." Then, these responses were computed using the same computation method from Studies 1

and 2 to provide an objective ambivalence index (*M* = 2.626, *SD* = 1.720). Similarly, a higher index score would mean more objective ambivalence towards receiving yearly flu vaccinations.

Initial Attitudes. The same attitude scale from Study 2 was adapted. Participants initial attitudes (*M* = 4.302, *SD* = 1.186, α = .898) toward receiving yearly flu vaccination were measured with the scale items presented in a randomized order. Their responses were recoded and then averaged, with higher score suggesting a more positive attitude.

Affective-Cognitive Focus. After reporting their attitudes, participants were induced with either an affective or a cognitive information focus using the same materials from Study 1. Thereafter, participants reported their subjective ambivalence.

Outcome Variable

Subjective Ambivalence. The same subjective ambivalence measure was used from Studies 1 and 2. Participants indicated their subjective ambivalence towards receiving yearly flu vaccination on three items. The responses were averaged, and higher score meant more ambivalence felt (*M* = 4.466, *SD* = 1.928, α = .942).

Results and Discussion

Table 5 shows the correlations among the predictors and the outcome variable.

Outcome Variable: Subjective Ambivalence

A hierarchical multiple regression was conducted with participants' subjective ambivalence as the outcome variable (see table 6). The analysis was

conducted in a similar way to that of Study 2.

Results revealed a main effect of initial attitudes toward flu vaccination on subjective ambivalence, B = -0.493, t(286) = -5.588, p < .001, 95% CI: [-0.666, -0.319], $r_{partial}$ = -.314. That is, participants with positive attitudes reported lower amount of subjective ambivalence. More objective ambivalence also meant more amount of subjective ambivalence, B = 0.345, t(286) = 5.679, p < .001, 95% CI: [0.225, 0.465], $r_{partial}$ = .318. However, no effect of affective-cognitive information focus was observed, B = -0.284, t(286) = -1.444, p = .150, 95% CI: [-0.672, 0.103], $r_{partial}$ = -.085.

These main effects were qualified by two significant two-way interactions. Firstly, the initial attitudes × affective-cognitive information focus interaction was significant, B = -0.467, t(283) = -2.614, p = .009, 95% CI: [-0.820, -0.115], $r_{partial}$ = -.154.[2] The other significant two-way interaction was that of the objective ambivalence × affective-cognitive information focus, B = -0.394, t(283) = -3.281, p = .001, 95% CI: [-0.630, -0.158], $r_{partial}$ = -.191.[3] Lastly, no interaction was observed between initial attitudes and objective ambivalence, B = 0.040, t(283) = 0.921, p = .358, 95% CI: [-0.045, 0.125], $r_{partial}$ = .055.

Of most importance, a significant three-way interaction was observed, B = 0.213, t(282) = 2.460, p = .014, 95% CI: [0.043, 0.383], $r_{partial}$ = .145. Again, to examine if initial attitudes mattered, this three-way interaction was decomposed by comparing individuals with positive or negative attitudes to the flu vaccination promotion information. Among participants with negative attitudes, which meant that they had inconsistent attitudes with the

information, a significant main effect was observed for objective ambivalence. That is, higher objective ambivalence meant more subjective ambivalence, $B = 0.630$, $t(282) = 5.354$, $p < .001$, 95% CI: [0.398, 0.861], $r_{partial} = .304$. There was no effect for affective-cognitive information focus, $B = 0.282$, $t(282) = 0.973$, $p = .331$, 95% CI: [-0.288, 0.852], $r_{partial} = .058$. Most importantly, a significant two-way interaction between objective ambivalence × affective-cognitive information focus was observed, $B = -0.672$, $t(282) = -4.095$, $p < .001$, 95% CI: [-0.995, -0.349], $r_{partial} = -.237$. This interaction suggested that for those with inconsistent attitudes, their intensity of objective ambivalence mattered in influencing the affective-cognitive matching effects on their self-reported subjective ambivalence. Specifically, for those with higher degree of objective ambivalence, a matched topic orientation and information focus led to even more subjective ambivalence compared to a mismatched topic orientation and information focus, $B = -0.875$, $t(282) = -2.356$, $p = .019$, 95% CI: [-1.605, -0.144], $r_{partial} = -.139$. On the other hand, for those with lower degree of objective ambivalence, a matched topic orientation and information focus led to even lesser subjective ambivalence compared to a mismatched topic orientation and information focus, $B = 1.438$, $t(282) = 3.305$, $p = .001$, 95% CI: [0.582, 2.294], $r_{partial} = .193$ (see figure 4a).

A different pattern of results was revealed for participants with positive attitudes, which meant that they had consistent attitudes with the information. For this group, higher degree of objective ambivalence meant more subjective ambivalence, $B = 0.433$, $t(282) = 3.939$, $p < .001$, 95% CI: [0.216, 0.649], $r_{partial} = .228$. In addition, there was a tendency for those with a cognitive focus to show more subjective ambivalence compared to those with a affective

focus, B = -0.542, $t(282)$ = -1.790, p = .075, 95% CI: [-1.139, 0.054], $r_{partial}$ = -.106. However, these main effects were not qualified by the two-way interaction of objective ambivalence × affective-cognitive information focus, B = -0.167, $t(282)$ = -1.107, p = .269, 95% CI: [-0.463, 0.130], $r_{partial}$ = -.066 (see figure 4b).

The main objectives of Study 3 were to demonstrate that the findings from Study 2 can be replicated in a cognitive topic and to give more confidence in the observed findings that considered the influence of initial attitudes. Importantly, the key findings from Study 2 did replicate for a cognitive topic in Study 3. That is, only for those with inconsistent attitudes to the information position but not those with consistent attitudes, a matched focus to the topic's orientation (i.e., cognitive focus to cognitive orientation) showed stronger correspondence between objective and subjective ambivalence compared to a mismatched focus (i.e., affective focus to cognitive orientation). Hence, the findings was replicated for both affective and cognitive topics.

In addition, across Studies 2 and 3, the obtained results aligned with the hypotheses derived from HSM. That is, only individuals who viewed the information to be inconsistent with their attitudes had their defense motivation triggered and thus, deliberated more over their conflicting evaluations to show the interaction pattern from Study 1. However, for those who viewed the information to be consistent, the lack of defense motivation to consider their conflicting evaluations led to no magnified effects of the objective ambivalence on subjective ambivalence despite a greater ability to retrieve both attitude-consistent and attitude-inconsistent information from having a

matched focus. Hence, HSM can be a useful model for guiding when a matched focus should be induced to strengthen or attenuate the tendencies for objective ambivalence to predict subjective ambivalence when considering one's attitudes and the position of the given information.

General Discussion

The main objective of this research was to examine the influences of matching affective-cognitive information focus with the topic's orientation on the objective-subjective ambivalence relationship. Importantly, the findings did support the hypothesized interaction effects. That is, under a matched focus, low objective ambivalent participants reported lower subjective ambivalence compared to those holding a mismatched focus. At the same time, those with high objective ambivalence reported greater subjective ambivalence under a matched focus compared to a mismatched focus (Study 1). Hence, a matched focus resulted in a stronger objective-subjective ambivalence correspondence compared to a mismatched focus.

Besides providing more evidence for Study 1, Studies 2 and 3 also revealed that the interaction effects depended on participants' initial attitudes which acted as a proxy to inform if their defense motivation would be triggered. That is, only those who perceived the information to be inconsistent to their attitudes were motivated to deliberate carefully over their evaluative conflicts, and thus, showed the interaction effects. Presumably, the attitude and information discrepancy activated the defense motivation in participants to scrutinize the information to protect their original position (Chaiken et al., 1989). This increased the saliency of their evaluative conflicts which then affected their subjective ambivalence. Conversely, those who viewed the

information to be consistent with their attitudes did not activate their defense motivation, and thus, showed no interaction effects (Chaiken et al., 1989; Giner-Sorolla & Chaken, 1997). That is, the lack of motivation to scrutinize their evaluative conflicts attenuated the magnification effects of having a matched focus on the objective-subjective ambivalence relationship.

Taken together, in relation to the information's position, the current findings support that initial attitudes play an important role as a proxy to inform whether having a matched focus would magnify the objective-subjective ambivalence relationship. Moreover, even though there were no significant attitude changes in participants after receiving the information,[4] the induced affective-cognitive focus still enhanced peoples' perceived ability and motivation to deliberate over their evaluations and reactions which then significantly influenced their subjective ambivalence. Hence, an implication of the current findings is that it is important to assess attitude strength outcomes such as subjective ambivalence, and not just attitudes, to get a comprehensive picture of the consequences of reading different types of information.

Another interesting preliminary finding is that the affective-cognitive matching effects were supported in both an affective and a cognitive topic that discouraged and promoted different health behaviors respectively. However, this should be interpreted with caution because there is only evidence for one affective and cognitive topic respectively. Future research may wish to address this by replicating with other affective and cognitive topics that discourage and promote behaviors respectively to provide more support.

Future research may also examine the extent to which the greater

subjective ambivalence that arises from an induced matched or mismatched focus would lead to greater receptivity to persuasion (Broemer, 2002; Salovey & Wegner, 2003). It may also provide additional approaches to enhance past ambivalence-related interventions to influence health behaviors (Armitage et al., 2003, Armitage & Conner, 2004; Conner et al., 2002; Wilson et al., 2012, Zhao & Cai, 2008). Moreover, since the affective-cognitive information focus can be induced via reading affective-cognitive information, it should be scalable to reach a wider audience to influence their objective-subjective ambivalence correspondence, thereby influencing additional downstream outcomes such as the correspondence between attitude and behavior (George et al., 2019; Nikoloudakis et al., 2018). As such, there is value to further understand and to replicate the findings from this approach in isolation or in conjunction with other techniques to better influence the strength with which people hold their evaluations.

Implications

Objective-Subjective Ambivalence Relationship. The present findings are consistent with past research showing a positive but modest objective-subjective ambivalence relationship (e.g., Priester & Petty, 1996). Moreover, it contributed a new approach to influence the correspondence between these two constructs via an affective-cognitive matching approach. This further attests to the idea that objective ambivalence alone cannot fully account for the conflict experience and that their correspondence can be influenced by additional factors (Haddock et al., 2017; Newby-Clarke et al., 2002; Weng & DeMarree, 2019). The affective-cognitive matching approach also provides a more nuanced take on how the increased saliency in evaluative conflicts can

have different influences on the objective-subjective ambivalence relationship. Specifically, it informs that the correspondence will be stronger under a matched information focus versus a mismatched information focus despite reading information in general would increase the saliency of the objective ambivalence. That is, when people have a clearer awareness of their positive and negative evaluations, stronger objective-subjective ambivalence correspondence is observed especially when they perceive it to be relatively easier to process and have greater accessibility to the information (Refling et al., 2013; Lee & Aaker, 2004).

Another implication involves further distinction of the components within evaluative conflicts. Specifically, since attitudes are contributed by both affective and cognitive components (Breckler, 1984; Crites et al., 1994; Zanna & Rempel, 1988), it is plausible to assume that the overall objective ambivalence can be further distinguished into intra-affective and intra-cognitive conflicts toward the same topic (Pacilli et al., 2012; Thompson et al., 1995). Since an affective-cognitive match between information focus and topic's orientation can influence the objective-subjective ambivalence correspondence, it hints at the possibility for greater influence on subjective ambivalence if the affective-cognitive matching approach also considered intra-affective and/or intra-cognitive conflicts. That is, this would increase the specificity of the matching parameters. Past research supports the possibility of intra-component conflicts as higher intra-cognitive conflicts were found to relate to lower attitude-behavior consistency for alcohol consumption (Armitage, 2003). However, it is not known if any studies have looked at intra-affective conflicts alone or in conjunction with intra-cognitive conflicts

yet. Nonetheless, this suggests that the respective affective-cognitive components of objective ambivalence may be another candidate to target for influencing subjective ambivalence using this approach. Hence, the consideration of these distinctions in the affective-cognitive matching approach would provide greater insights toward changing subjective ambivalence, and consequently, the attitudinal outcomes affected by subjective ambivalence.

***Affective-Cognitive Matching*.** The present findings are also consistent with past research that argued for the applicability of the distinction between affect and cognition basis in attitudes to other variables such as information focus and topic's orientation (Bondur et al., 2000; See et al., 2008; Zanna & Rempel, 1988). More importantly, the findings also support that matching effects result in stronger outcomes compared to mismatching effects even when matched to the topic's orientation (e.g., Haddock et al., 2008; Keer et al., 2013). In addition, it also expands the outcomes influenced by affective-cognitive matching effects to include subjective ambivalence. Hence, this may demonstrate the potential utility of extending the matching effects to target the normatively affective-cognitive preferences of groups and even cultures for better attitudinal outcomes (Cervellon & Dubé, 2002). This would be especially useful when individual differences in affective-cognitive preferences are not known or are difficult to obtain.

In situations when individual differences can be examined, the current findings also suggest the possibility of influencing the objective-subjective ambivalence correspondence by matching to the dispositional factors such as individuals' affective-cognitive meta-bases or the need for affect/cognition.

Past research showed that affective-cognitive focus matched to one's dominant meta-bases or need for affect/cognition resulted in greater attitudinal change compared to a mismatched focus (e.g. Haddock et al., 2008; Keer et al., 2013; See et al., 2008). Moreover, individuals' meta-bases also signal the processing preference for information with an affective or a cognitive orientation (See, 2018). Hence, matching to these dispositional factors may either reinforce the effects found when matched to the topic's orientation or may even work regardless of the topic's orientation. Taken together, the affective-cognitive matching approach has the flexibility to influence the objective-subjective ambivalence relationship by tailoring to a wide range of factors (e.g., topic's orientation, information focus, dispositional factors).

Generalizing to Other Attitudinal Strength Measures. Given the usefulness of the affective-cognitive matching approach with attitudinal strength dimensions such as ambivalence and certainty in influencing attitudinal outcomes, it is likely that this approach would also work with other attitude strength dimensions such as attitude clarity and correctness (Petrocelli et al., 2007). For example, just as a matched information focus can influence low ambivalent participants to feel even less conflicted, it should also influence people with high attitude clarity and correctness to show stronger attitudes if they repeatedly express opinions or receive supporting consensus information with a matched focus compared to a mismatched focus. Likewise, the reverse may also happen such that repeating strong arguments or receiving opposing consensus information with a matched focus would give greater attitude change than a mismatched focus. Hence, the potential to use the affective-cognitive matching approach across different attitudinal strength

dimensions should be examined in greater detail to provide greater accuracy in matching or mismatching the affective-cognitive information focus to the topic's orientation to influence the attitudinal strength factor(s) of interest.

In addition, since an affective-cognitive matching approach can influence numerous attitudinal strength dimensions, it also hints at the potential to understand how these strength factors can work together in an affective-cognitive matching framework. Although past research has examined the interactive influences of ambivalence and certainty on attitudinal outcomes such as attitude stability (Luttrell et al., 2016a), it has not yet considered how focusing on affect or cognition might provide a more nuanced influence on the targeted outcomes. That is, since attitudinal strength dimensions can work interactively to affect attitudinal outcomes, it would be interesting to examine if affective-cognitive matching effects would only influence specific or all attitudinal strength factors simultaneously. Thus, this would provide a more detailed understanding of the complexity regarding attitudinal strengths' influences on important attitudinal outcomes.

Limitations and Future Directions

One limitation in this research is that only defense motivation, a well-established factor for increased deliberation, was examined (Giner-Sorolla & Chaiken, 1997; Liberman & Chaiken, 1992). However, other alternative factors may also give similar outcomes. For example, within HSM, impression motives may hinder deliberations when a person tries to hold similar position to the person or the group that they value (Chaiken et al., 1996). Specifically, under the influence of impression motives, individuals may show biased instead of careful information processing or discount inconsistent thoughts in

order to match the positions of the valued person or group. However, those without such motives would likely be relatively more critical of any inconsistencies in their information processing or recalled thoughts. Hence, with impression motives, individuals' inconsistencies about the topic should be less salient compared to those without impression motives. As such, it is expected that those under the influence of impression motives would show relatively weaker objective-subjective ambivalence relationship as they are not motivated to deliberate carefully. Conversely, those without impression motives should show relatively stronger objective-subjective ambivalence relationship especially if they have greater processing fluency from a matched topic orientation and information focus. Future research can thus examine other potential factors that influence information processing to see if they give similar result patterns as this research. Nonetheless, regardless of the specific factor that triggered the different degree of information processing, the relatively greater deliberations are likely to be the proximal factor that accounts for this observed affective-cognitive matching pattern.

Another limitation is that the psychological mechanisms which account for the interaction effects on subjective ambivalence from the affective-cognitive matching approach are not directly measured. Specifically, these mechanisms are postulated to be the greater subjective processing fluency and the greater recall and accessibility of conflicting reactions. In addition, other mechanisms could also account for observed patterns as well, such as a greater self-relevance that motivates careful information processing under a matched focus compared to a mismatched focus (e.g., DeBono & Packer, 1991; Petty & Cacioppo, 1979). Hence, to definitively confirm the mechanisms for the

effects, future studies should include and examine the mediating and moderating mechanisms such as the subjective processing fluency, recall and accessibility of conflicting reactions, and the perceived position that the information takes (Clark et al., 2008b; Mayer & Tormala, 2010; Petty & Cacioppo, 1977). The inclusion of these measures should help confirm and explain the observed effects and allow for the exploration of potential competing explanations. Nonetheless, regardless of the mechanism, the current research provides support for the hypothesis that a matched focus can magnify the relationship between objective ambivalence and subjective ambivalence, and thus warrants further investigation of such magnifying effects of matching.

Also, the downstream consequences of subjective ambivalence were not examined. While much research has shown that subjective ambivalence can influence many important outcomes (Clark et al., 2008a; DeMarree et al., 2014; van Harreveld et al., 2009), it is unclear if the subjective ambivalence elicited using the current approach would have similar effects compared to naturally arising ones and those elicited via other methods. Moreover, some attitudinal outcomes have also taken an affective-cognitive distinction such as the requested advocacy versus spontaneous advocacy which was found to be better predicted by cognitive and affective meta-bases respectively (Teeny & Petty, 2018). Hence, subjective ambivalence that is influenced by a matched affective or cognitive information focus may also show differential effects on these outcomes. Thus, future studies may include both traditional and newer attitudinal outcomes to confirm the utility of subjective ambivalence elicited via an affective-cognitive matching approach.

Finally, the affective or cognitive orientation of the topics was considered separately across studies instead of within a single study. That is, there could be some inherent differences across the topics selected that accounted for the interaction patterns observed. Hence, the affective-cognitive matching effects might apply to only some topics, or that the interaction patterns are actually attributed to some inherent properties of the selected topics that are not identified or examined in the current set of studies. As such, in future research, it would be valuable to manipulate the affective-cognitive orientation of the topic so that they can be examined concurrently using the same topic in the same study. This would further add to the robustness of the affective-cognitive matching approach as well.

Beyond Affective-Cognitive Matching. Future research can examine if the affective-cognitive matching approach could be extended to other domains such as values and morality. Past research has found that topics can vary by the values it represents (e.g., charity represents altruism) and whether it has a morality element (e.g., meat consumption) to it (Berndsen & van der Pligt, 2005; Maio & Olson, 2000). In addition, past research has also found that information can be crafted to elicit a focus on values and morality (Maio & Olson, 2000; Luttrell et al., 2016b). As such, these variants of information focus would likely also be useful in eliciting a matching effect similar to the ones observed in the affective-cognitive matching approach. Thus, it would be interesting for future studies to examine if a value focus or a moral focus would be more effective for affecting the objective-subjective ambivalence correspondence in value-driven or moral-driven topics such as capital punishment or meat consumption compared to an affective or a cognitive

information focus.

Cross-Cultural Differences. Lastly, another interesting question that could be asked is whether these affective-cognitive matching effects would be identical across cultures. Past research has suggested that Asians prefer having an emotional information focus while Westerners prefer a cognitive information focus (e.g., Lin, 2001), and these differences have been explained by the concept of high- and low-context cultures (Hall, 1976). That is, Eastern cultures are high-context cultures that prefer information with implied meaning such as those that elicit emotional focus, while Western cultures are low-context cultures that favor direct information such as those that elicit a focus on rationality and logic (Choi et al., 2005; Hall, 1976; Mueller, 1987). Hence, there could be a difference in the required intensity of the affective-cognitive focus across cultures to replicate the patterns found in the current findings.

Moreover, since past research has shown that East Asians are more comfortable with holding conflicting thoughts compared to their Western counterparts (Peng & Nisbett, 1999), the observed pattern in this study might be stronger and easier to elicit in individuals from Western cultures. This could also be due to East Asians perceiving the experienced conflicts as a cue for being thoughtful about the attitudinal target, and thus being more tolerant of the discomfort (Ng et al., 2012). Put differently, East Asians might require even stronger affective-cognitive evaluative conflicts to view their conflicting experiences as uncomfortable instead of being considerate. Hence, future studies may examine this via both observational and experimental designs using participants with broader characteristics (e.g., age, occupation) that are

more equivalent to have greater confidence in the cross-cultural generalizability of the affective-cognitive matching effect.

Conclusion

Extant literature has revealed that the objective-subjective ambivalence correspondence can vary in strength depending on other factors, and thus hold implications for many downstream outcomes (e.g., attitude-behavior consistency, information processing). This present research sought to investigate such factors and added a new approach to influence the objective-subjective ambivalence relationship via an affective-cognitive matching approach. Demonstrating the utility of affective-cognitive matching, a match between the affective-cognitive information focus and the topic's orientation was found to interact with participants' objective ambivalence to magnify their subjective ambivalence experience. That is, matching leads to a stronger objective-subjective ambivalence relationship than mismatching. Moreover, this occurs mainly when a person's attitudes are inconsistent with the received information, thereby triggering a defense motivation to deliberate over their evaluations and reactions. Future research can focus on exploring and confirming the mediating and moderating mechanisms for the current findings by examining both affective and cognitive topics in the same study through the manipulation of the topic's orientation. It would also be useful to replicate the findings for more affective and cognitive topics. Thus, the current findings are a promising extension of our understanding of the affective-cognitive matching and the degree of objective-subjective ambivalence correspondence.

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Footnotes

1. This interaction revealed that the degree of attitude positivity towards drinking sweetened drinks mattered in objective-subjective ambivalence relationship strength. For those with positive attitudes, more objective ambivalence meant more subjective ambivalence, B = 0.315, $t(265)$ = 3.594, $p < .001$, 95% CI: [0.142, 0.487], $r_{partial}$ = .216. This was stronger among those with negative attitudes, B = 0.834, $t(265)$ = 10.274, $p < .001$, 95% CI: [0.674, 0.994], $r_{partial}$ = .534.
2. This interaction revealed that degree of attitude positivity toward receiving flu vaccination mattered for the subjective ambivalence experience. For those with positive attitudes, which meant that they had inconsistent attitudes with the information, a matched topic orientation and information focus resulted in more subjective ambivalence compared to those with a mismatched topic orientation and information focus, B = -0.816, $t(283)$ = -2.871, p = .004, 95% CI: [-1.376, -0.256], $r_{partial}$ = -.168. However, a matched or mismatched focus did not affect those with negative attitudes (i.e., those with consistent attitudes with the information), B = 0.293, $t(286)$ = 1.003, p = .317, 95% CI: [-0.282, 0.868], $r_{partial}$ = .059.
3. This interaction revealed a similar pattern to Study 1. That is, for those with a higher degree of objective ambivalence, a matched topic orientation and information focus resulted in more subjective ambivalence compared to those with a mismatched topic orientation and information focus, B = -0.939, $t(283)$ = -3.306, p = .001, 95% CI: [-1.499, -0.380], $r_{partial}$ = -.193. Whereas for those with a lower degree of objective ambivalence, having a matched or a mismatched focus did not significantly affect their subjective

ambivalence experience. However, the trend was in the predicted direction where a matched topic orientation and information focus showed lower subjective ambivalence compared to those with a mismatched topic orientation and information focus, B = 0.416, $t(283)$ = 1.465, p = .144, 95% CI: [-0.143, 0.976], $r_{partial}$ = .087.

4. In order to rule out attitudes as an alternative explanation for the effects on subjective ambivalence, we also measured participants' attitudes after they received the information. Across three studies, none of the highest order interaction effects were significant for post-information attitudes (p > .05). Specifically, the objective ambivalence × affective-cognitive information focus interaction was not significant for post-information attitudes in study 1, B = -0.014, $t(132)$ = -0.102, p = .919, 95% CI: [-0.290, 0.261], $r$partial = -.009, with the initial attitudes and post-information attitudes being M = 4.515, SD = 1.424 and 4.680, SD = 1.302. Likewise in study 2, the three-way interaction was not significant, B = -0.043, $t(264)$ = -0.999, p = .318, 95% CI: [-0.128, 0.042], $r$partial = -.061, with the initial attitudes being M = 3.934, SD = 1.165 and the post-information attitudes being M = 3.741, SD = 1.192. Study 3 showed no three-way interaction as well, B = 0.070, $t(282)$ = 1.677, p = .095, 95% CI: [-0.012, 0.151], $r$partial = .099, with the initial attitudes being M = 4.622, SD = 1.248 and the post-information attitudes being M = 4.320, SD = 1.186. Thus, the observed effects on subjective ambivalence were not due to participants' changing their attitudes from the flu vaccination or sweetened drink information.

Table 1

Correlations among variables for study 1

	1	2	3
1. Information Focus	---	.189*	-.025
2. Objective Ambivalence		---	.092
3. Subjective Ambivalence			---

*$p < .05$ **$p < .01$

Table 2

Regression analysis for subjective ambivalence for study 1

Predictor	*B*	Partial *r*	R^2_{Change}
Step 1			
Objective Ambivalence	.119	.098	
Affective-Cognitive Information Focus	-.173	-.043	.010
Step 2			
Objective Ambivalence × Affective-Cognitive Information Focus	-.690**	-.285	.080

Note. R^2 = .090, *p < .05 **p < .01 ***p < .001

Table 3

Correlations among variables for study 2

	1	2	3	4
1. Information Focus	---	-.040	.003	-.027
2. Objective Ambivalence		---	.669**	.333**
3. Subjective Ambivalence			---	.309**
4. Initial Attitudes				---

*$p < .05$ **$p < .01$

Table 4

Regression analysis for subjective ambivalence for study 2

Predictor	*B*	Partial *r*	R^2_{Change}
Step 1			
Initial Attitudes	.197*	.124	
Objective Ambivalence	.645***	.632	.456
Affective-Cognitive Information Focus	.147	.042	
Step 2			
Initial attitudes × Objective Ambivalence	-.223***	-.286	
Initial attitudes × Affective-Cognitive Information Focus	-.320	-.103	.047
Objective Ambivalence × Affective-Cognitive Information Focus	.077	.050	
Step 3			
Initial attitudes × Objective Ambivalence × Affective-Cognitive Information Focus	.243**	.161	.013

Note. R^2 = .516, *p < .05 **p < .01 ***p < .001

Table 5

Correlations among variables for study 3

		1	2	3	4
1.	Information Focus	---	-.023	-.073	-.025
2.	Objective Ambivalence		---	.411**	-.334**
3.	Subjective Ambivalence			---	-.404**
4.	Initial Attitudes				---

*$p < .05$ **$p < .01$

Table 6

Regression analysis for subjective ambivalence for study 3

Predictor	*B*	Partial *r*	R^2_{Change}
Step 1			
Initial Attitudes	-.493***	-.314	
Objective Ambivalence	.345***	.318	.254
Affective-Cognitive Information Focus	-.284	-.085	
Step 2			
Initial attitudes × Objective Ambivalence	.040	.055	
Initial attitudes × Affective-Cognitive Information Focus	-.467**	-.154	.035
Objective Ambivalence × Affective-Cognitive Information Focus	-.394**	-.191	
Step 3			
Initial attitudes × Objective Ambivalence × Affective-Information Cognitive Focus	.213*	.145	.015

Note. R^2 = .304, *p < .05 **p < .01 ***p < .001

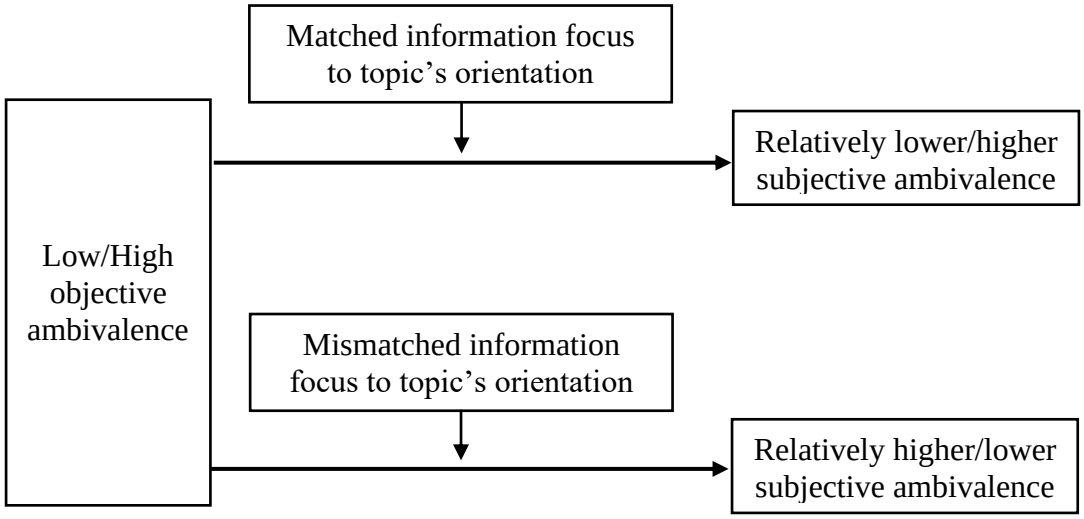


Figure 1. Stronger objective-subjective ambivalence relationship is expected when there is a matched affective-cognitive focus to the topic's orientation compared to a mismatched focus.

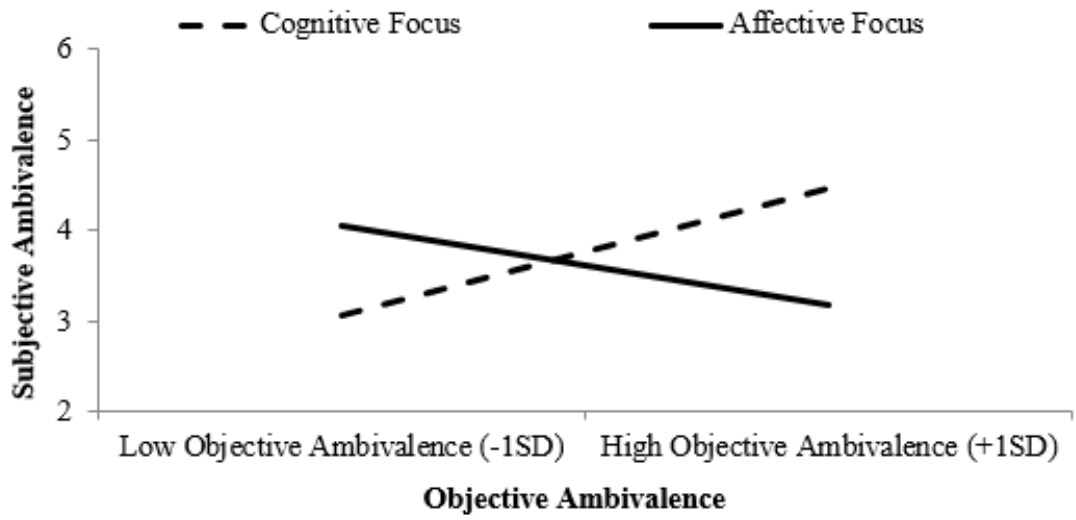


Figure 2. A matched cognitive focus to the cognitive orientation of flu vaccination showed a stronger objective-subjective ambivalence relationship compared to a mismatched affective focus.

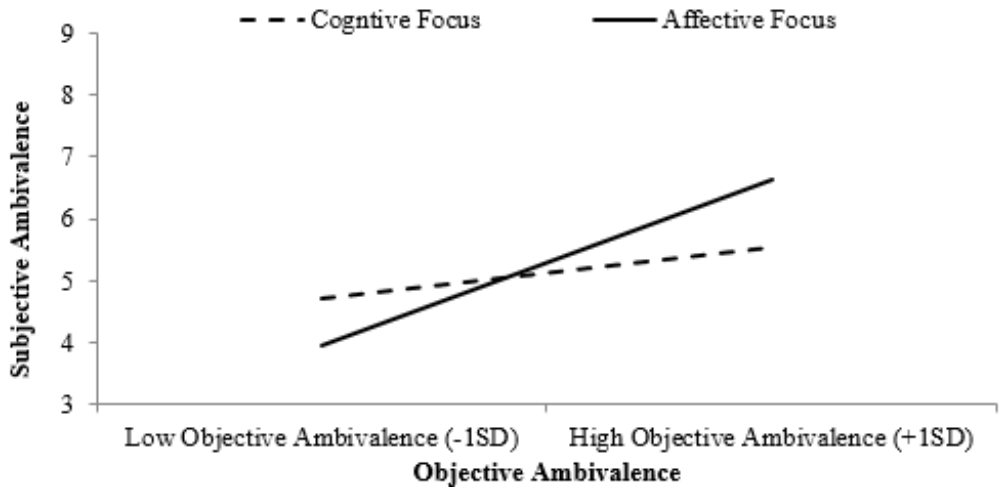


Figure 3a. Participants with inconsistent attitudes to the information showed stronger objective-subjective ambivalence relationship from a matched affective focus to the affective orientation of sweetened drinks compared to a mismatched cognitive focus.

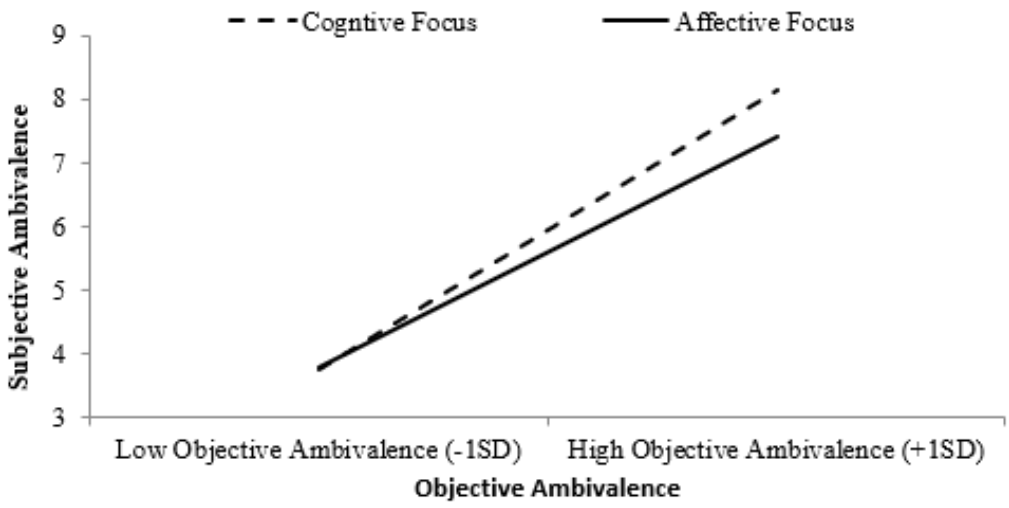


Figure 3b. Participants with consistent attitudes to the information showed similar strength in their objective-subjective ambivalence relationship in both a matched and mismatched information focus to the affective orientation of sweetened drinks.

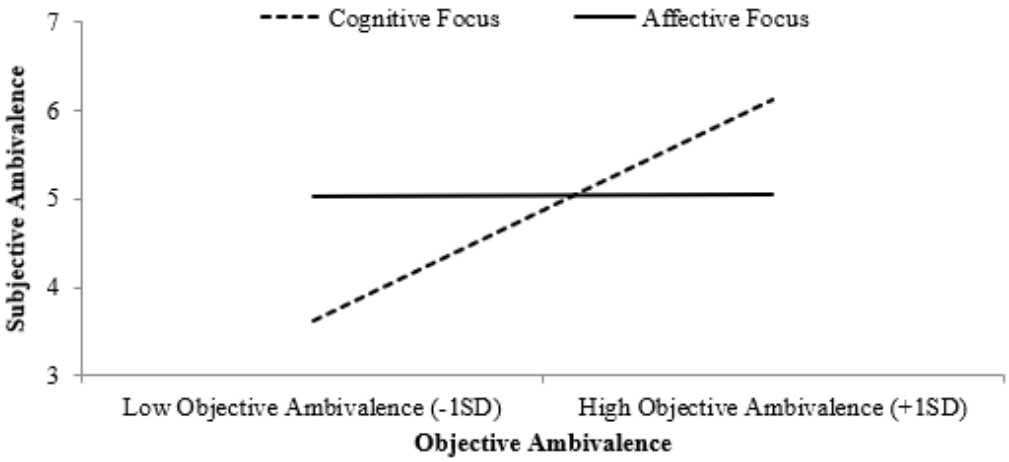


Figure 4a. Participants with inconsistent attitudes to the information showed stronger objective-subjective ambivalence relationship from a matched cognitive focus to the cognitive orientation of flu vaccination compared to a mismatched affective focus.

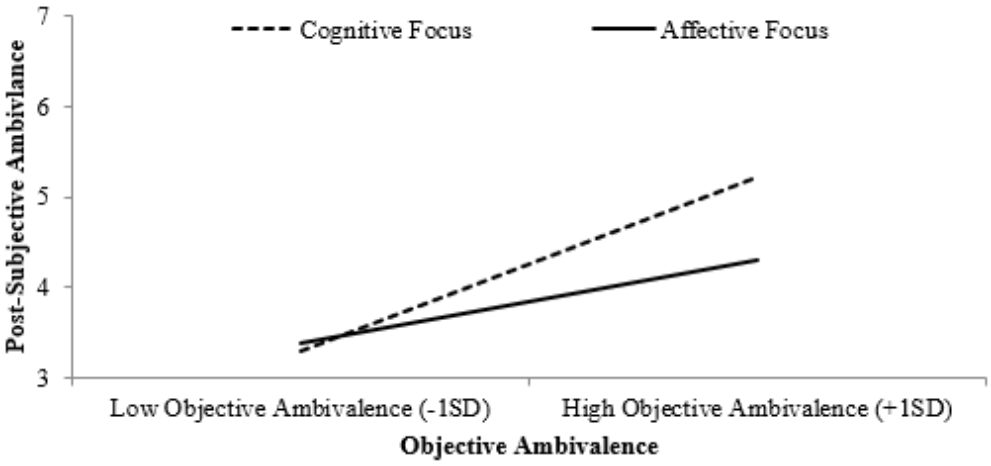


Figure 4b. Participants with consistent attitudes to the information showed similar strength in their objective-subjective ambivalence relationship in both a matched and mismatched information focus to the cognitive orientation of flu vaccination.

Appendix A

Objective Ambivalence Questionnaire

Please indicate the extent to which you agree with each of the following statements.

Strongly disagree 1 2 3 4 5 6 7

Strongly agree

Study 1

1. I feel happy about receiving yearly flu vaccination.
2. I feel bored about receiving yearly flu vaccination.
3. It can feel delightful to receive yearly flu vaccination.
4. It can feel sad to receive yearly flu vaccination.
5. It feels exciting to receive yearly flu vaccination.
6. It feels disgusting to receive yearly flu vaccination.

Study 2

1. Drinking sweetened drinks is valuable for your health.
2. Drinking sweetened drinks creates a lot of challenges.
3. It is unnecessary to drink sweetened drinks.
4. It is wholesome to drink sweetened drinks.
5. Drinking sweetened drinks is beneficial.
6. Drinking sweetened drinks is useless if you do not improve other areas of your health.

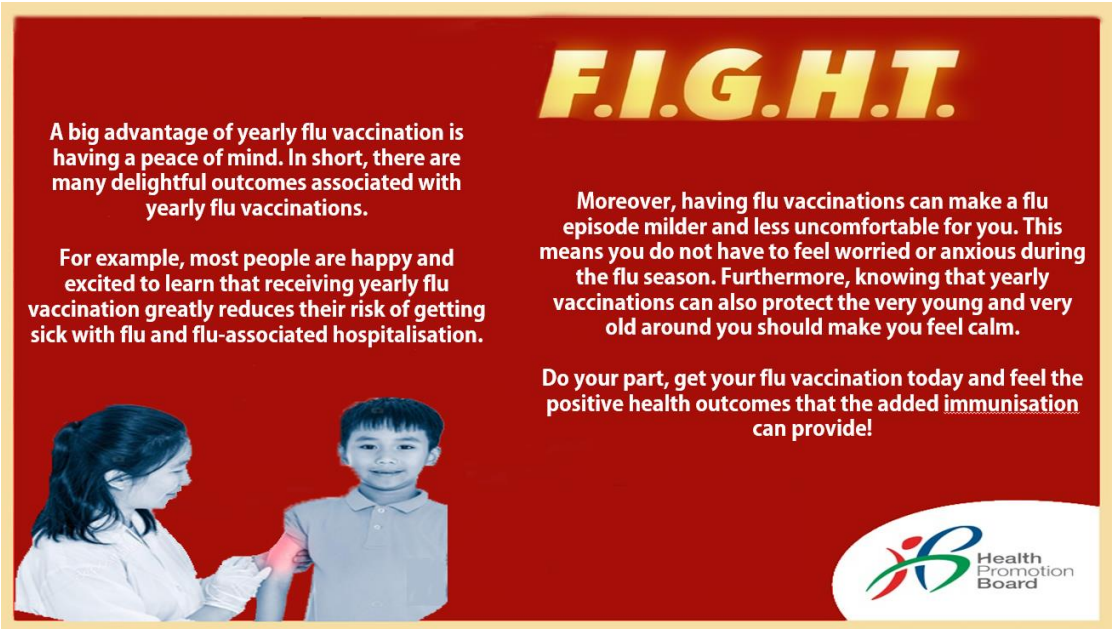
Study 3

1. I believe it is wise to get flu vaccinations.
2. I believe it is unnecessary to get flu vaccinations.
3. Going for flu vaccination is *valuable* for your health.
4. Going for flu vaccination *is useless* if you do not improve other areas of your health.
5. Maintaining immunity against flu is beneficial.
6. Going for flu vaccination takes a long time and creates a lot of hassle.

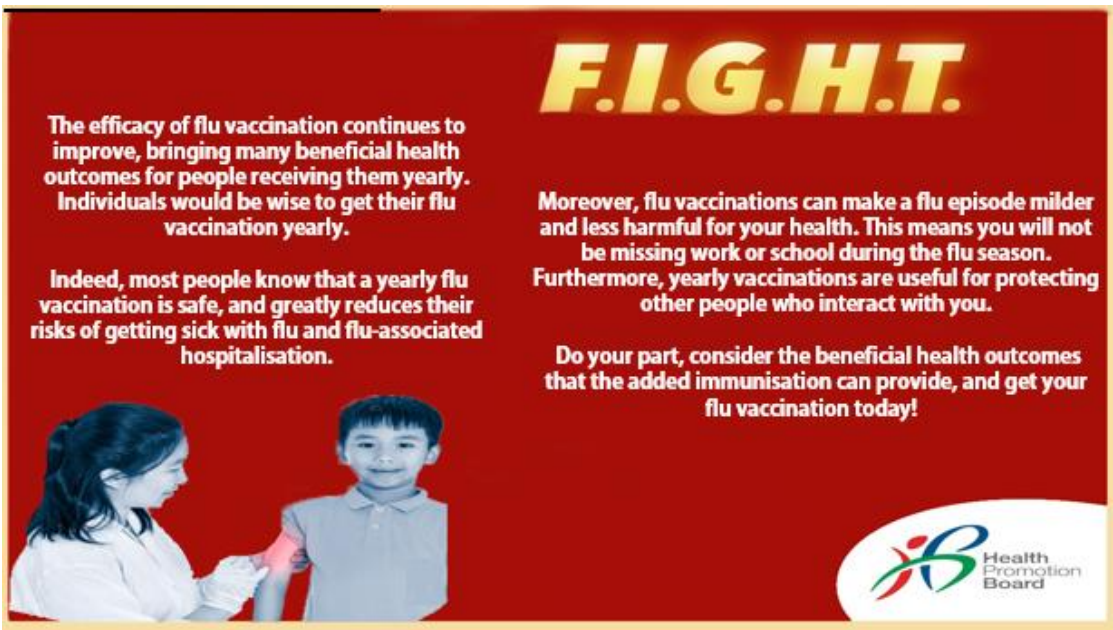
Appendix B

Information on Promoting Receiving Flu Vaccination

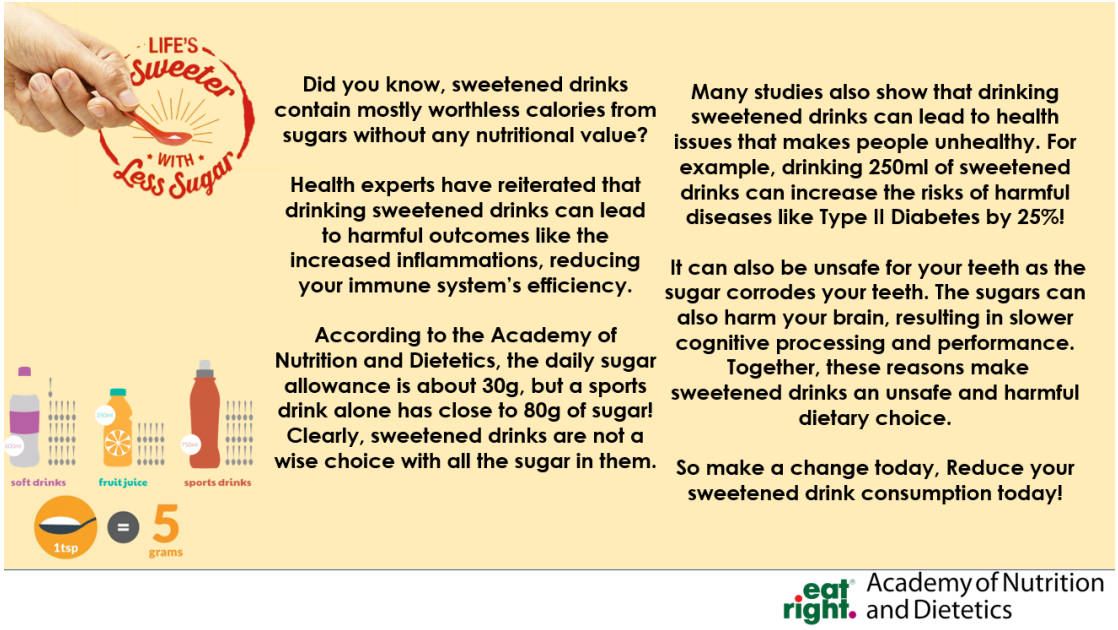
Affect-Focused



Cognition-Focused



Information on Reducing Sweetened Drinks Consumption

Affect-Focused

Cognitive-Focused